

AI-Powered Food Waste Reduction Assistant for College Canteens

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SDG Alignment

- **SDG 12:** Responsible Consumption and Production
- **Secondary SDG:** SDG 2: Zero Hunger

1.Abstract:

Food waste in college canteens is a major challenge that leads to unnecessary resource consumption and environmental impacts. This project proposes an AI-powered assistant that predicts the amount of food required based on historical demand and provides recommendations to minimize food wastage. The system uses machine learning to estimate food demand and helps canteen managers optimize food preparation. By reducing waste, the project promotes sustainable consumption and efficient resource utilization.

2.Problem Statement:

How might we use AI to predict food demand in college canteens so that food wastage can be reduced and resources can be used more sustainably?.

Target Users

- College canteen managers
- Hostel mess administrators
- Students
- Educational institutions

Current Gaps

- Food is prepared based on assumptions.
- Excess food is often wasted.
- No prediction system exists.
- Resource consumption increases.

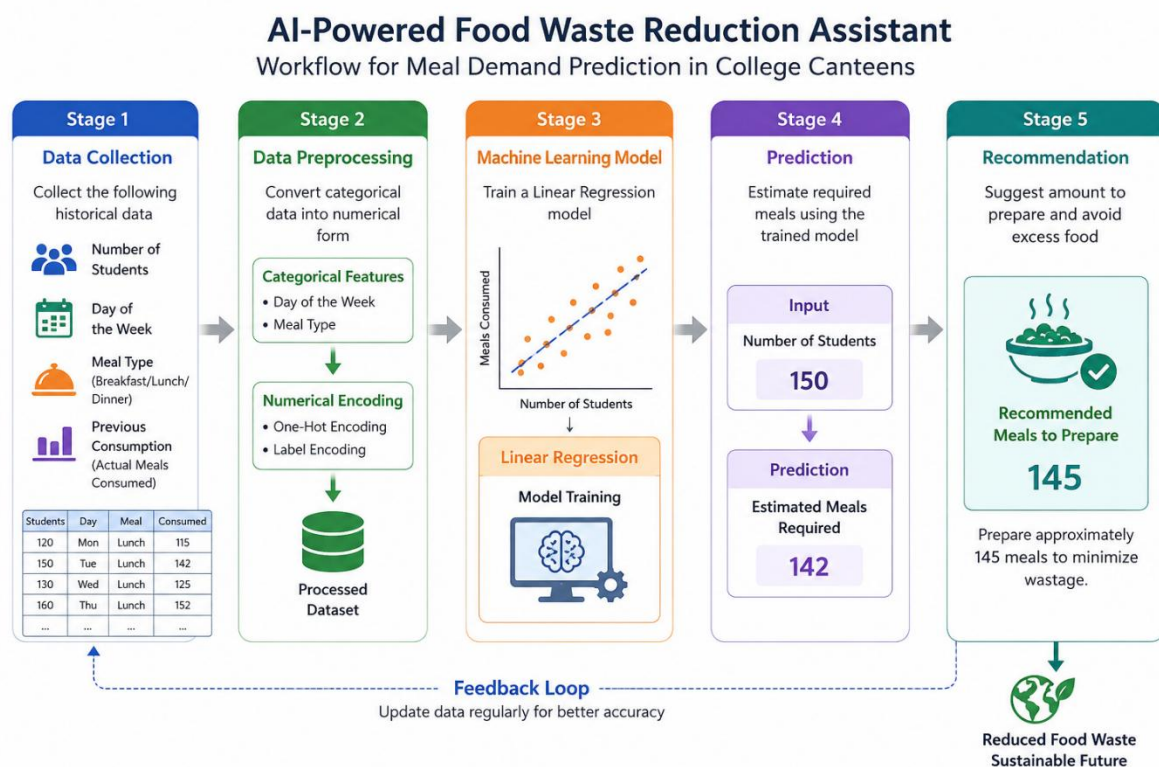
3.AI Solution Overview:

The system predicts the quantity of food required using historical attendance and meal consumption data.

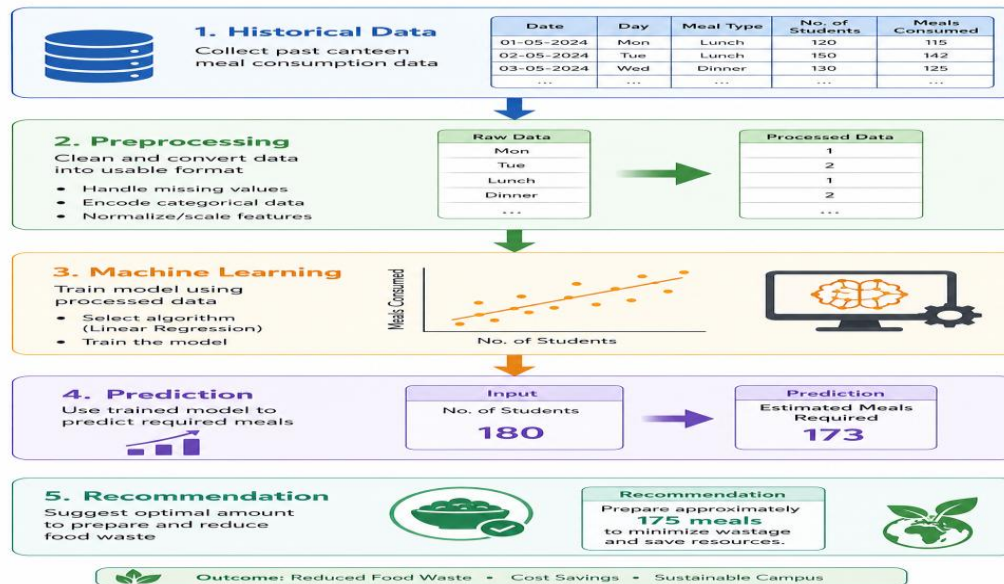
AI Role:

- ✓ Prediction
- ✓ Decision Support
- ✓ Pattern Detection

4.Methodology:



5.Architectre:



6.Required Python Code:

food_waste_prediction.py

```
import pandas as pd

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder

# Historical data
data = {
    'Students_Present': [100, 120, 110, 140, 130, 150],
    'Day': ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat'],
    'Meal_Type': ['Breakfast', 'Lunch', 'Dinner', 'Lunch', 'Breakfast', 'Dinner'],
    'Previous_Consumption': [95, 115, 108, 135, 125, 145],
    'Meals_Consumed': [98, 118, 110, 138, 128, 148]
```

```
}  
  
# Create dataframe  
df = pd.DataFrame(data)  
  
# Encode categorical columns  
day_encoder = LabelEncoder()  
meal_encoder = LabelEncoder()  
  
df['Day'] = day_encoder.fit_transform(df['Day'])  
df['Meal_Type'] = meal_encoder.fit_transform(df['Meal_Type'])  
  
# Features and target  
X = df[['Students_Present', 'Day', 'Meal_Type', 'Previous_Consumption']]  
y = df['Meals_Consumed']  
  
# Train model  
model = LinearRegression()  
model.fit(X, y)  
  
# User input  
students = int(input("Enter number of students present: "))  
day = input("Enter day (Mon/Tue/Wed/Thu/Fri/Sat): ")  
meal = input("Enter meal type (Breakfast/Lunch/Dinner): ")  
prev_consumption = int(input("Enter previous meal consumption: "))  
  
# Convert inputs to numerical values  
day_num = day_encoder.transform([day])[0]  
meal_num = meal_encoder.transform([meal])[0]  
  
# Predict meals required  
prediction = model.predict([[students, day_num, meal_num, prev_consumption]])
```

```
print("\nPredicted meals required =", round(prediction[0]))  
# Recommendationrecommended = round(prediction[0] + 5)  
print("Recommended meals to prepare =", recommended)_meals[0]))
```

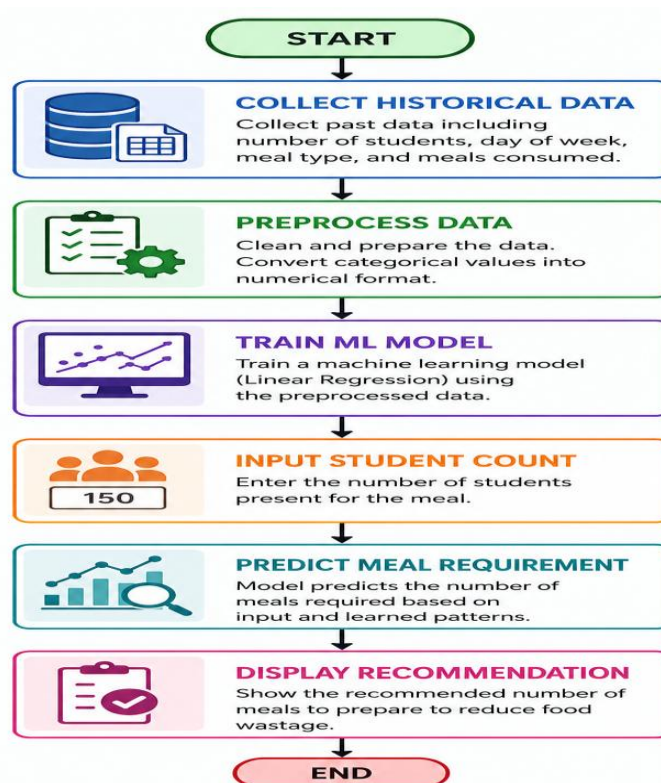
Sample Input

Enter number of students present: 180
Enter day (Mon/Tue/Wed/Thu/Fri/Sat): Mon
Enter meal type (Breakfast/Lunch/Dinner): Lunch
Enter previous meal consumption: 170

Output

Predicted meals required = 174
Recommended meals to prepare = 179

7.Flowchart:



8.Responsible AI Considerations:

Fairness

Predictions should not favor any particular group.

Transparency

Users can understand how predictions are generated.

Ethics

No harmful use of data.

Privacy

No personal information of students is collected.

9.Expected Impact:

- Reduce food wastage by 20–30%.
- Lower operational costs.
- Promote sustainable consumption.
- Support SDG 12 goals.

10.Demo Idea:

Input

Students Present = 180

AI Prediction

Meals Required = 173

Recommendation

Prepare approximately 175 meals
to minimize wastage.

11.Impact Statement:

If implemented, this system can help colleges reduce food waste, save resources, lower costs, and contribute toward sustainable development. It provides an AI-driven approach for making smarter decisions in food preparation and resource management.

12.Conclusion:

The **AI-Powered Food Waste Reduction Assistant for College Canteens** provides a smart and sustainable approach to minimizing food wastage by predicting meal requirements based on historical consumption patterns, student attendance, day of the week, and meal type. By leveraging machine learning, the system helps canteen managers make informed decisions about food preparation, reducing excess waste and optimizing resource utilization. This project promotes responsible consumption, lowers operational costs, and contributes to **SDG 12: Responsible Consumption and Production**, demonstrating how AI can be used responsibly to create a positive environmental and social impact.